

Accessibility

different disabilities

testing, legal requirements

user interface features

curb cuts

Abilities

- How are your abilities different from other people?
- How will your abilities change in the future?
- How do your abilities change in different environments?

Abilities

- The “average person” is just a statistical ideal
 - No one person is “average”
- Individual performance and capabilities vary significantly
- Each one of us has permanent (e.g., Loss of two legs) and/or temporary disabilities
 - arise due to nature of our environment or our health
 - What forms of temporary disabilities are there?

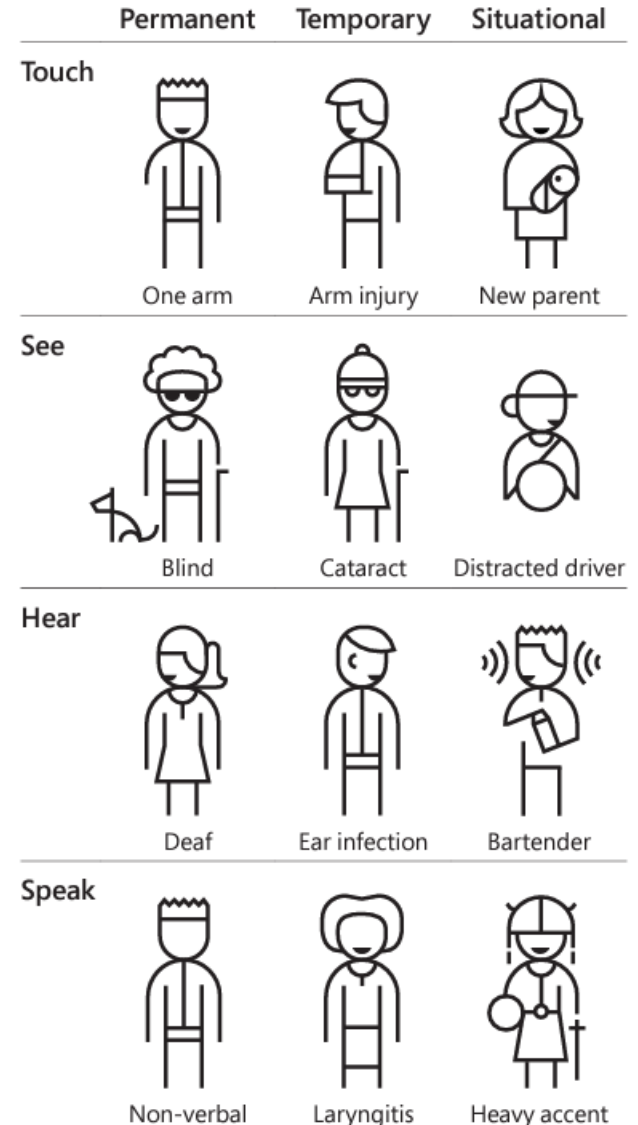
Temporary Disabilities

- Sick, injured
 - Temporarily impaired cognitive capabilities
 - Temporary loss of motor capabilities
- Driving a car
 - Limited attentional bandwidth
- Making dinner at home while attending to children
 - Limited attentional bandwidth
- Underwater diving
 - Impaired sight, hearing, mobility



The Persona Spectrum

- We use the Persona Spectrum to Understand related mismatches and motivations across a **spectrum of permanent, temporary, and situational** scenarios.
- It's a quick tool to help foster empathy and to show how a solution scales to a broader audience.



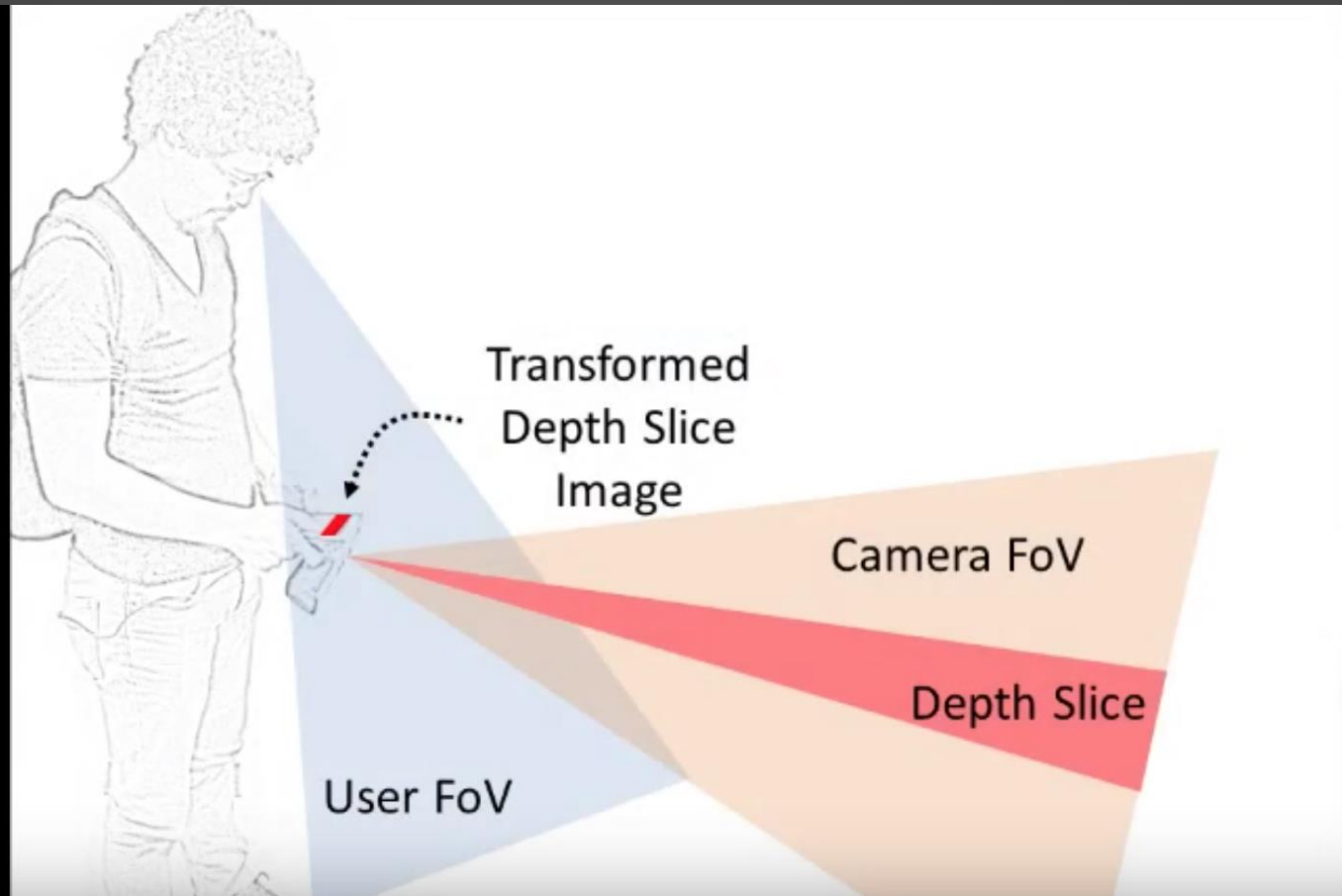
Padded lampposts for distracted texters being tested in London

BY JOSHUA TOPOLSKY MARCH 6TH 2008, AT 5:39:00 AM ET

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According to a recent report, human beings are becoming so incredibly stupid that they require cushioned lamppost bases so that when they run into them they don't mess up their



CrashAlert (Hincapié-Ramos and Irani, 2013)

- https://youtu.be/HN_F-coZqt8?t=44s

Tapping Performance when Walking

- Task: tap with stylus on targets of varying sizes and distances
- Conditions:
 - Sitting
 - Treadmill: slow
 - Treadmill: fast
 - Obstacle course (self-paced)
- Measures:
 - Pointing speed
 - Errors



Fig. 1. Demonstration of someone walking on the treadmill while doing target selection task.

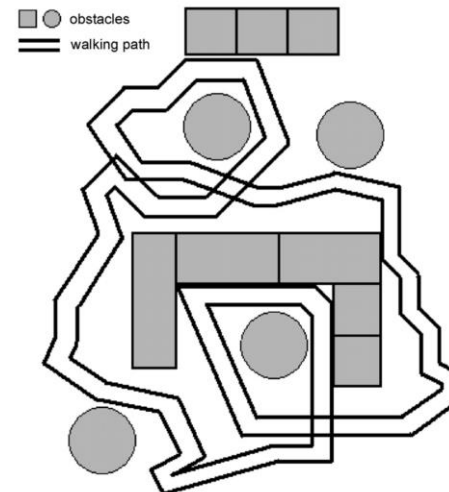
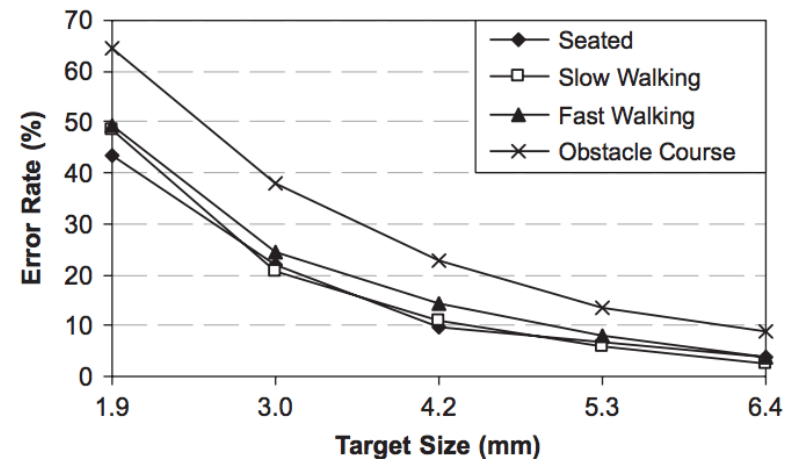
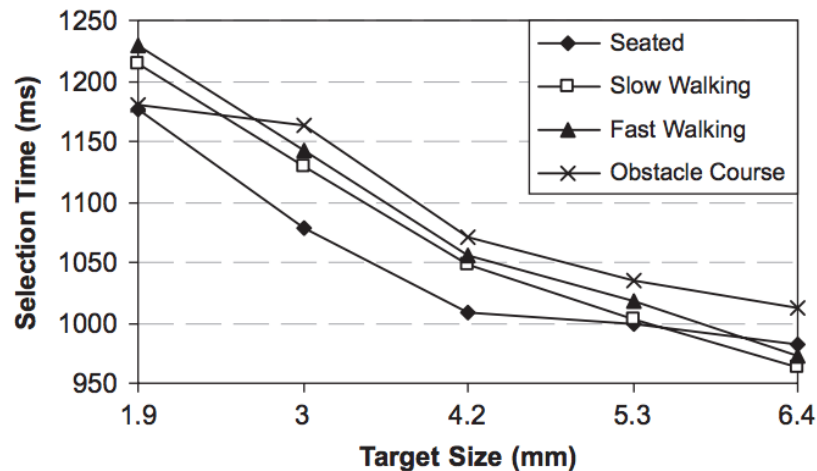


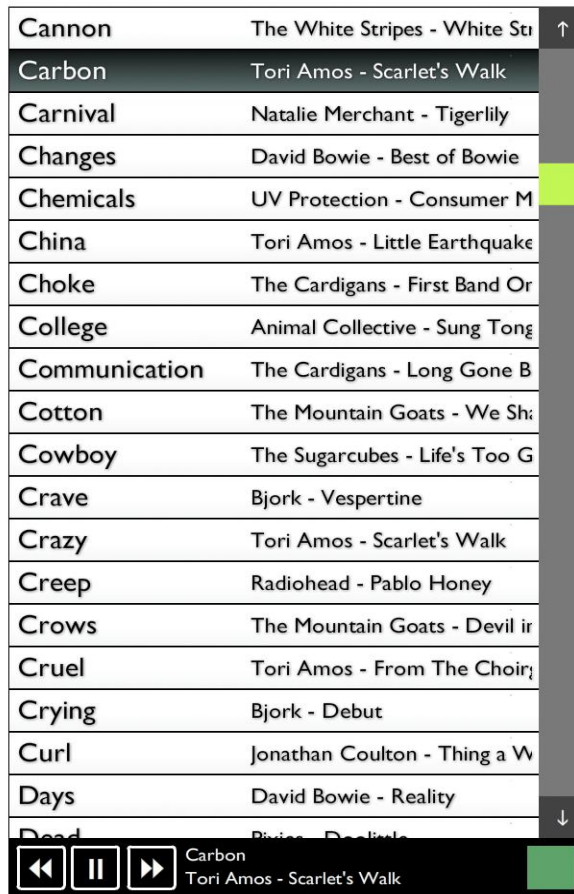
Fig. 2. Diagram of the obstacle course path used to create the attention-intensive walking condition.

Lin et al. How do people tap when walking? An empirical investigation of nomadic data entry. *International Journal of Human-Computer Studies* pp. 759-769

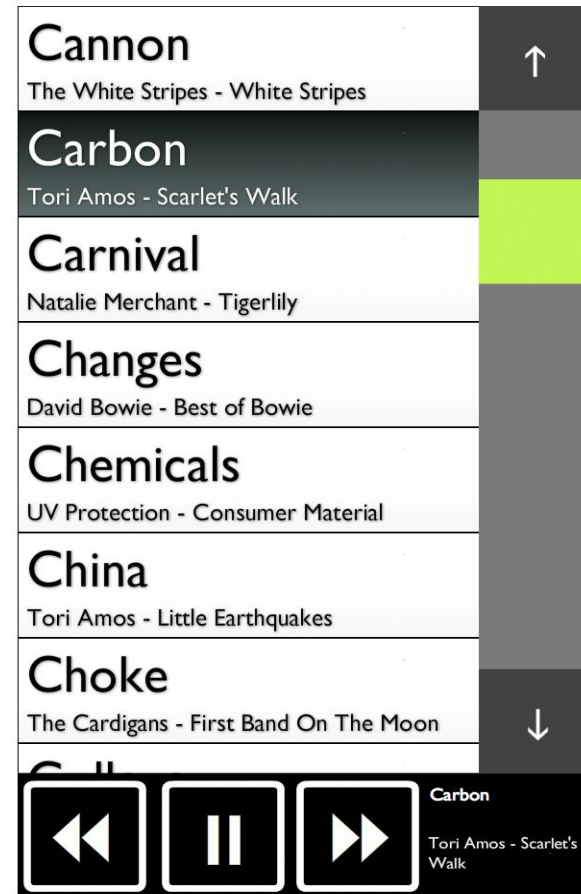
Tapping Performance when Walking

- The time to select individual targets did not differ significantly between conditions
- Error rates increased dramatically when the participants walked through the obstacle course even they reduced their walking speed by 36%.
- 90% or higher accuracy for targets with diameters as small as 4.2, 5.3, and 6.4 mm under the seated, treadmill, and the obstacle course conditions, respectively





Sitting UI

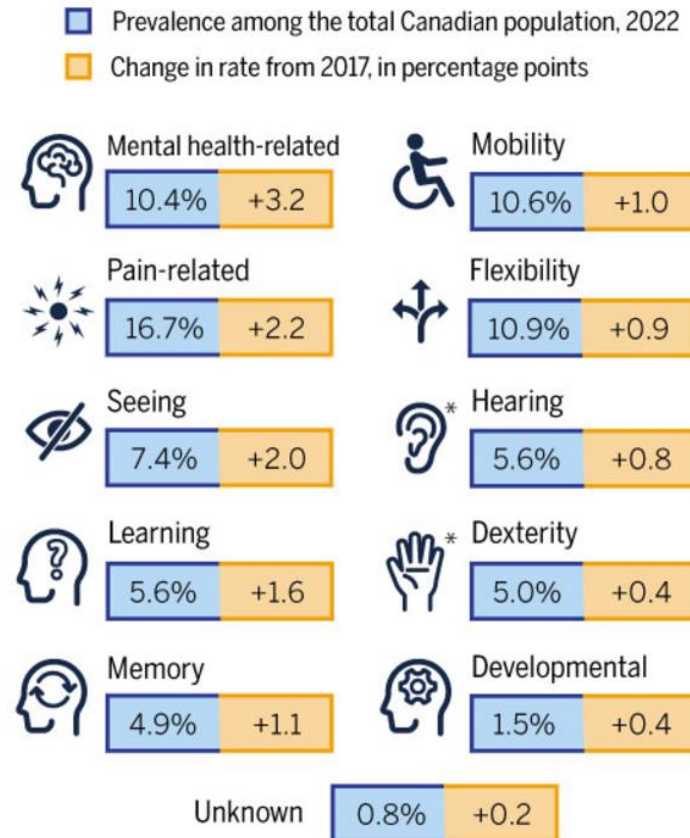


Walking UI

Kane et al. Getting off the treadmill: evaluating walking user interfaces for mobile devices in public spaces. MobileHCI '08: Proceedings of the 10th international conference on Human computer interaction with mobile devices and services (2008)

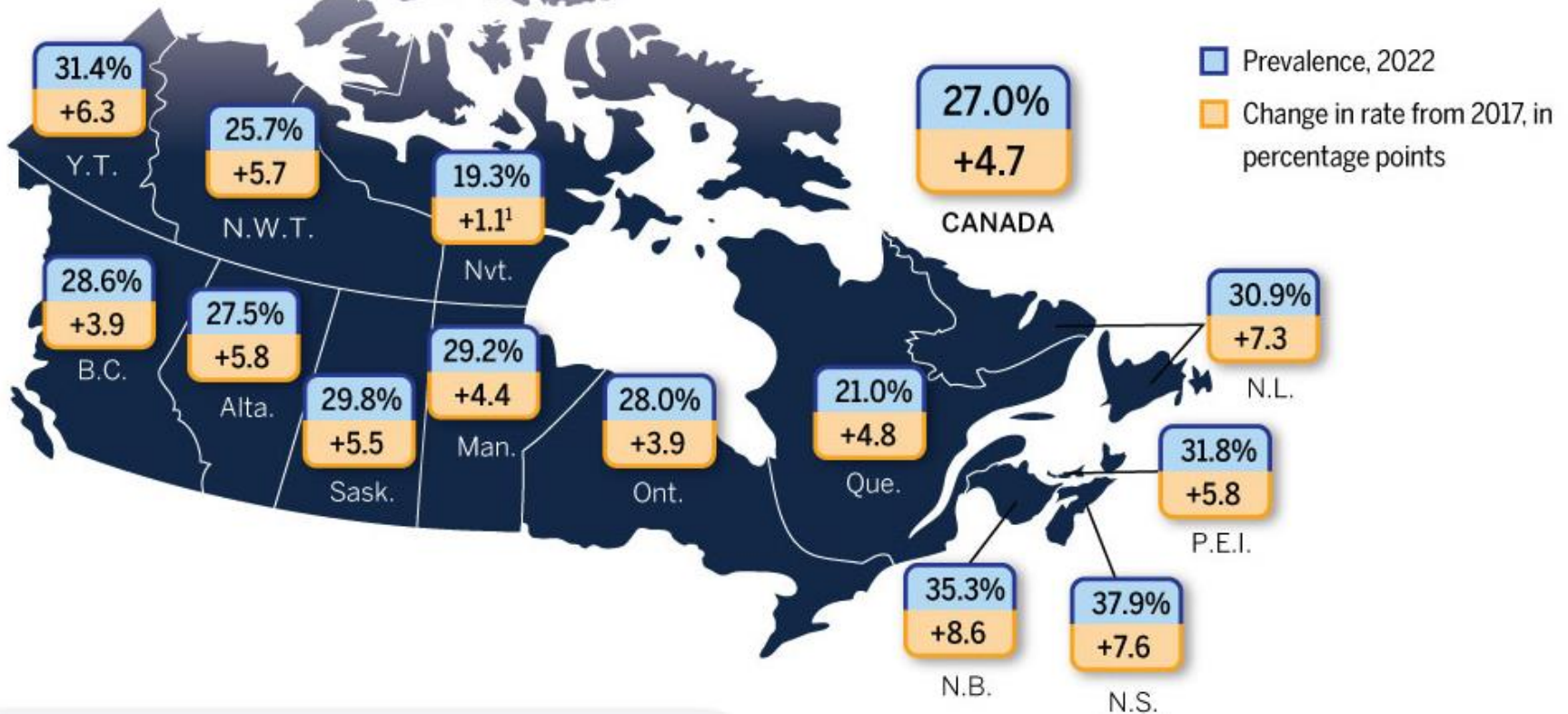
People with Disabilities

- In 2022, 27.0% of Canadians aged 15 and over, or 8.0 million people, had at least one disability.
- Mental health-related, pain-related, and seeing disabilities had the largest increases

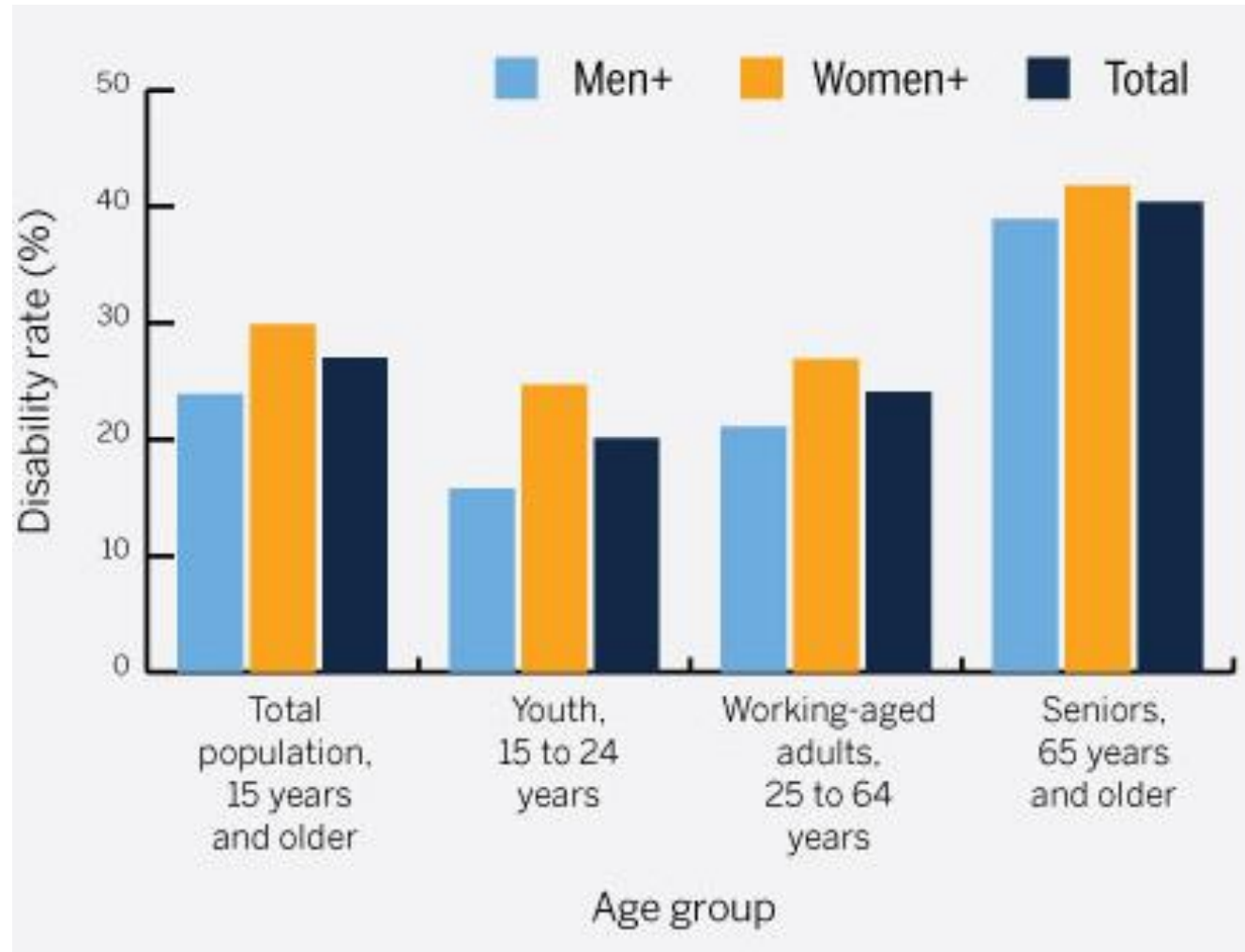


People with Disabilities

From 2017 to 2022, the disability rate rose in all provinces and most of the territories

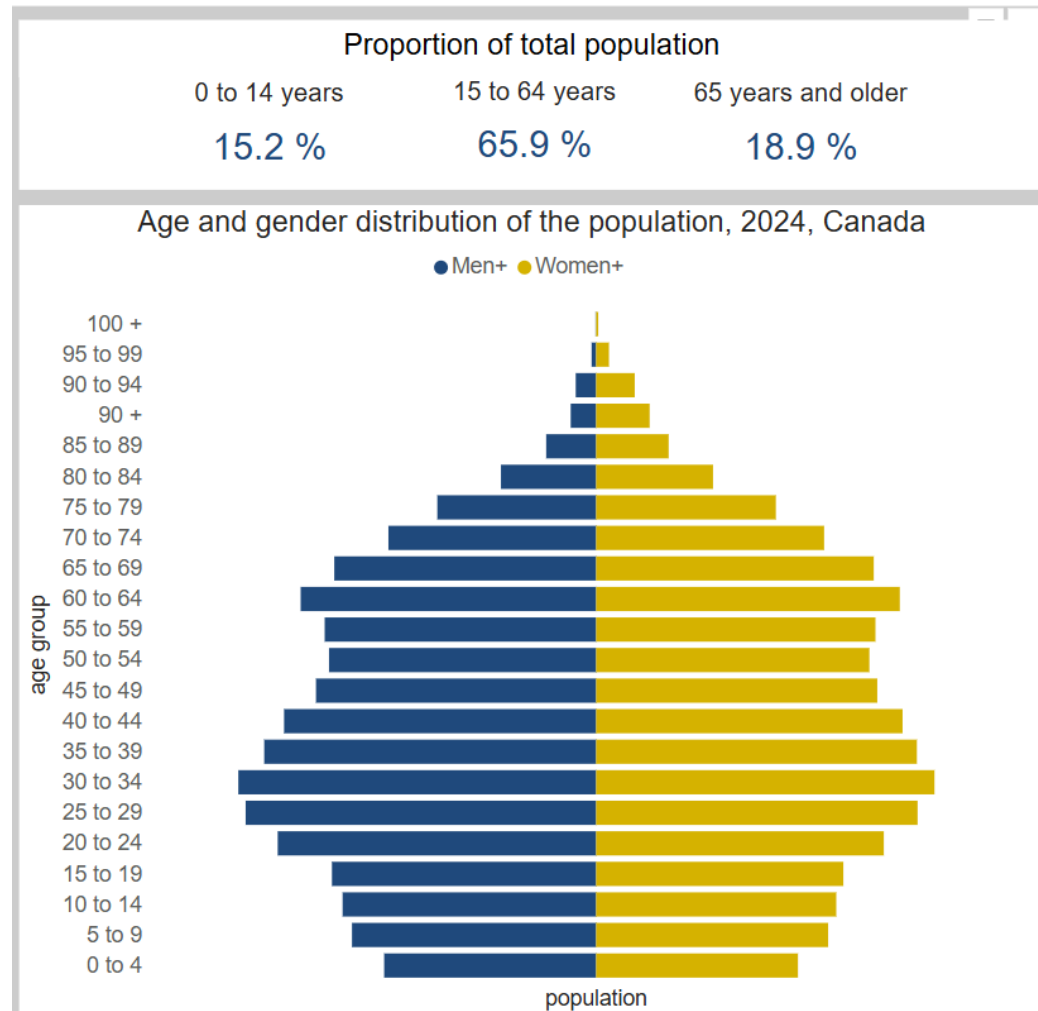


People with Disabilities



Aging Population

- Canadian seniors now outnumber children



Age-Related Disabilities

- Decreased visual acuity
- Problems with low levels of illumination
- Reduced ability to distinguish colors
- Decreased contrast sensitivity
- Visual search is harder, takes longer
- Pattern recognition more difficult
- Tracking and processing moving targets harder
- Reduced spatial abilities
- Increased difficulty with motor movements
- Decline in auditory function



MIT AgeLab's Age Gain Now Empathy System

- <https://youtu.be/czuww9rp5f4>

The Key Areas for Considering

- **Visual:** Long-sightedness, blindness, color blindness, are all forms of visual disability you need to cater for in your design.
- **Motor/Mobility:** This category doesn't just extend to problems with the use of the hands and arms, but also with other muscular or skeletal conditions.
- **Auditory:** Auditory disabilities affect the hearing and come in varying degrees of severity, up to and including total deafness.
- **Seizures:** Some individuals can be affected by light, motion, flickering, etc. on screen, thus triggering seizures.
- **Learning:** It's also important to remember that not all disabilities are physical. Learning and cognitive disabilities can also influence accessibility.

Some Suggestions

- If you use a CMS, choose one that **supports accessibility standards**. Drupal and WordPress, for example, support these
- Use **alt text** on your images; if you use images to enhance content, then a screen reader will need to explain them— that's what the alt text is for
- Have a **link strategy**. Instead of “Click here”. “Read more about the UBC, at their website.”
- Choose **colors carefully**; if in doubt, test your color schemes with some color-blind people
- **Don't refer just to the color**; “click the red button” might not be helpful to a color-blind person
- More information: <https://www.interaction-design.org/literature/article/accessibility-usability-for-all>

WAVE: Web Accessibility Evaluation Tool

<https://wave.webaim.org/report#/https://ok.ubc.ca/>

The screenshot displays the WAVE (Web Accessibility Evaluation Tool) interface, powered by WebAIM, evaluating the website <https://ok.ubc.ca/>. The tool's summary panel on the left indicates 11 errors, 2 contrast errors, 20 alerts, 27 features, 83 structural elements, and 38 ARIA issues. The main panel shows the University of British Columbia Okanagan Campus website with various accessibility error overlays, including tabindex issues and a missing alt text for an image. The website header includes the UBC logo and navigation links. The footer features a large image of three people smiling on a bus.

WAVE powered by WebAIM
web accessibility evaluation tool

Address: <https://ok.ubc.ca/>

Styles: OFF ☒ ON

Summary

- Summary
- Details
- Reference
- Order
- Structure
- Contrast

11 Errors

2 Contrast Errors

20 Alerts

27 Features

83 Structural Elements

38 ARIA

[View details](#)

The following apply to the entire page:

- *en-US*

THE UNIVERSITY OF BRITISH COLUMBIA
Okanagan Campus

Navigation: About, Admissions, Student Life, Academics, Research, Apply to UBC, Parents, Alumni, Community, Give UBC

Prospective Students | Current Students | Faculty and Staff

WAVE: Web Accessibility Evaluation Tool

Address: https://ok.ubc.ca/

Styles: OFF ☐ ON ☒

Details

Summary
 Details
 Reference
 Order
 Structure
 Contrast

11 Errors

- 9 X Linked image missing alternative text
- 1 X Missing form label
- 1 X Broken skip link

Okanagan Campus

3333 University Way
Kelowna, BC Canada V1V 1V7
Tel 250.807.8000

[Campus Map](#) | [Parking Info](#) | [Mail Info](#)

About UBCO

[Facts](#) | [History and Milestones](#) | [Leadership](#)
[Careers](#) | [Library](#) | [Give to UBCO](#)
[IT Services](#) | [Webmail](#) | [Website Feedback](#)

Faculties and Schools

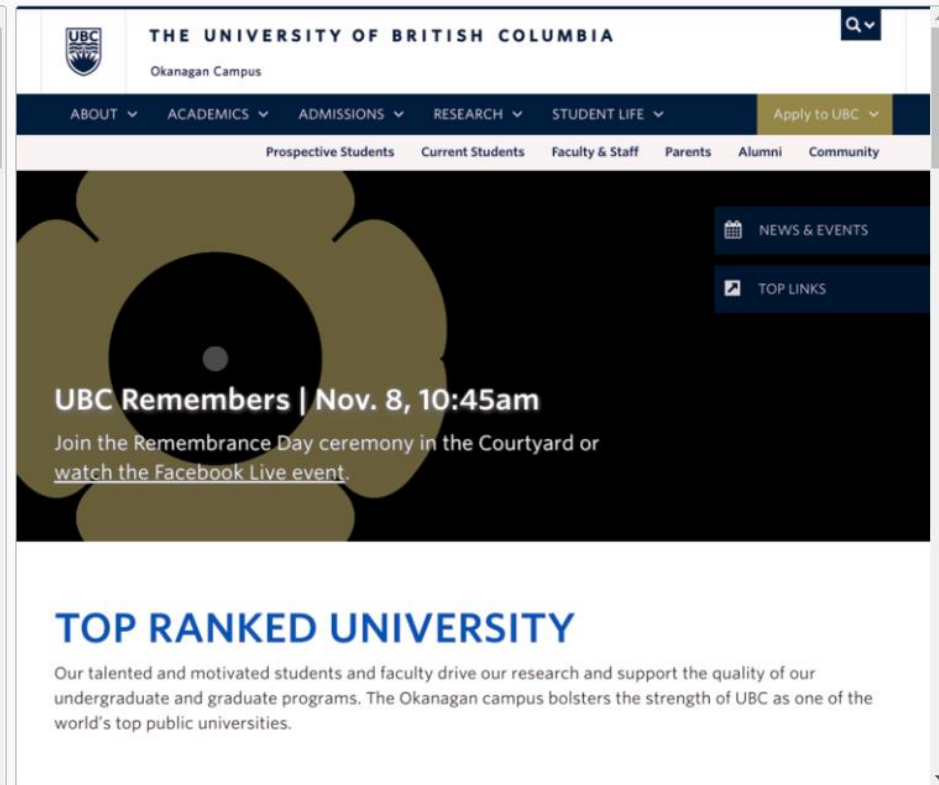
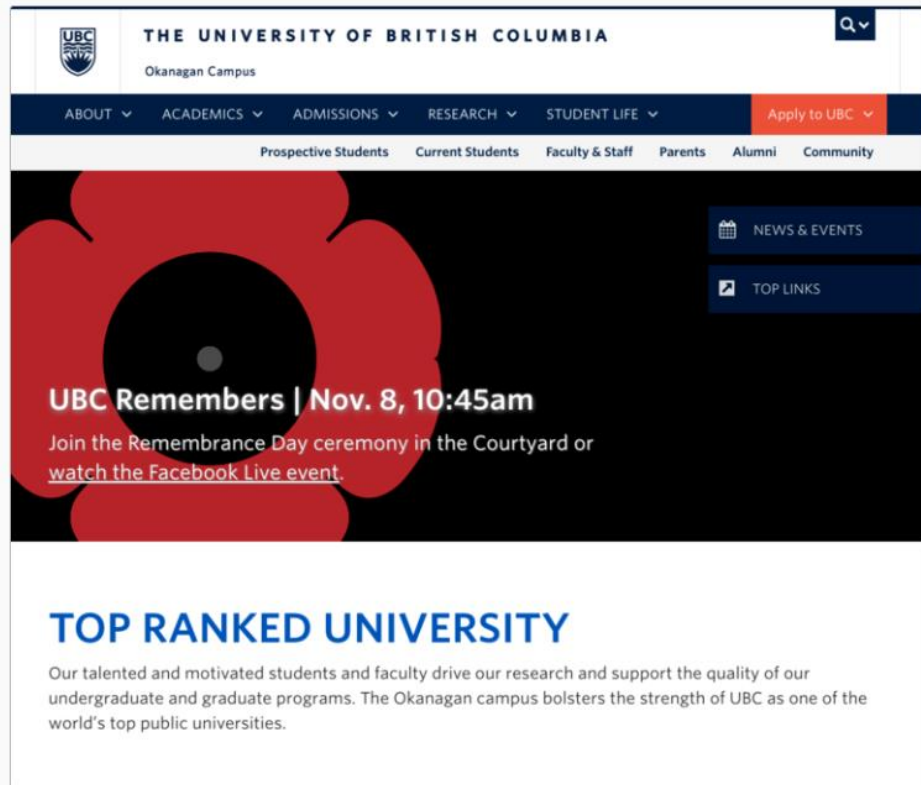
[Arts and Social Sciences](#)
[Creative and Critical Studies](#)
[Education](#)
[Engineering \(Applied Science\)](#)
[Health and Social Development Management](#)
[Medicine \(Southern Medical\) Science](#)
[College of Graduate Studies](#)

Okanagan Campus

Prospective Students

Colourblind Web Page Filter

<https://www.toptal.com/designers/colorfilter>



Designing for Disabilities

- Why design for users with disabilities?
 - It's the ethical thing to do
 - Because it's the law in some countries
 - Because of the “curb cut phenomenon”

Legal Obligations

- The United Nations and the World Wide Web Consortium (W3C) have declarations and guidelines on ensuring that everyone can get access to information that is delivered through software technologies
 - <https://www.w3.org/WAI/fundamentals/accessibility-intro/>
- Canada
 - Web Standards for the Government of Canada often used
<http://www.tbs-sct.gc.ca/pol/doc-eng.aspx?id=23601>
- Ontario
 - How to make websites accessible
 - <https://www.ontario.ca/page/how-make-websites-accessible>
- United States' Disabilities and Rehabilitation Acts (Section 508)
 - All government facilities, services, and communications must be accessible to individuals with disabilities

Lawsuit over web site accessibility for the blind becomes class action

By [Nate Anderson](#) | Last updated October 3, 2007 1:34 PM

A lawsuit brought in 2006 by a [blind student at the University of California-Berkeley](#) has now morphed into a [class action case against US retailer Target](#). A federal judge has just certified a nationwide class in the case, which alleges that Target's web site is not fully accessible to the blind. It's a case that could help establish the ways in which the Americans With Disabilities Act applies to the Internet, and it has already generated a ruling that, in California at least, commercial web sites must be accessible.

The case focuses on the alleged lack of descriptive "alt" tags in Target's HTML, making the site difficult to navigate with screen reading software. The use of image maps is also claimed to make the site inaccessible.

Public locations in the real world have long been required to abide by the ADA, but the law was written in the days before the Web, and it remains unclear how it should be applied to web sites. One of the lawyers from Disability Rights Advocates, which is handling the case, sees inaccessibility as a simple issue of discrimination, online or off.

"Target Corporation has led a battle against blind consumers in a key area of modern life: the Internet economy," said Larry Paradis in a statement after the ruling. "The court's decision today makes clear that people with disabilities no longer can be treated as second-class citizens in any sphere of mainstream life."

In addition to certifying the case as a national class action under the ADA, Judge

Solve for one, extend to many

- Designing for people with permanent disabilities can seem like a significant constraint, but the resulting designs can actually benefit a much larger number of people.
- Example:
 - a device designed for a person who has one arm could be used just as effectively by a person with a temporary wrist injury or a new parent holding an infant



One arm



Arm injury



New parent

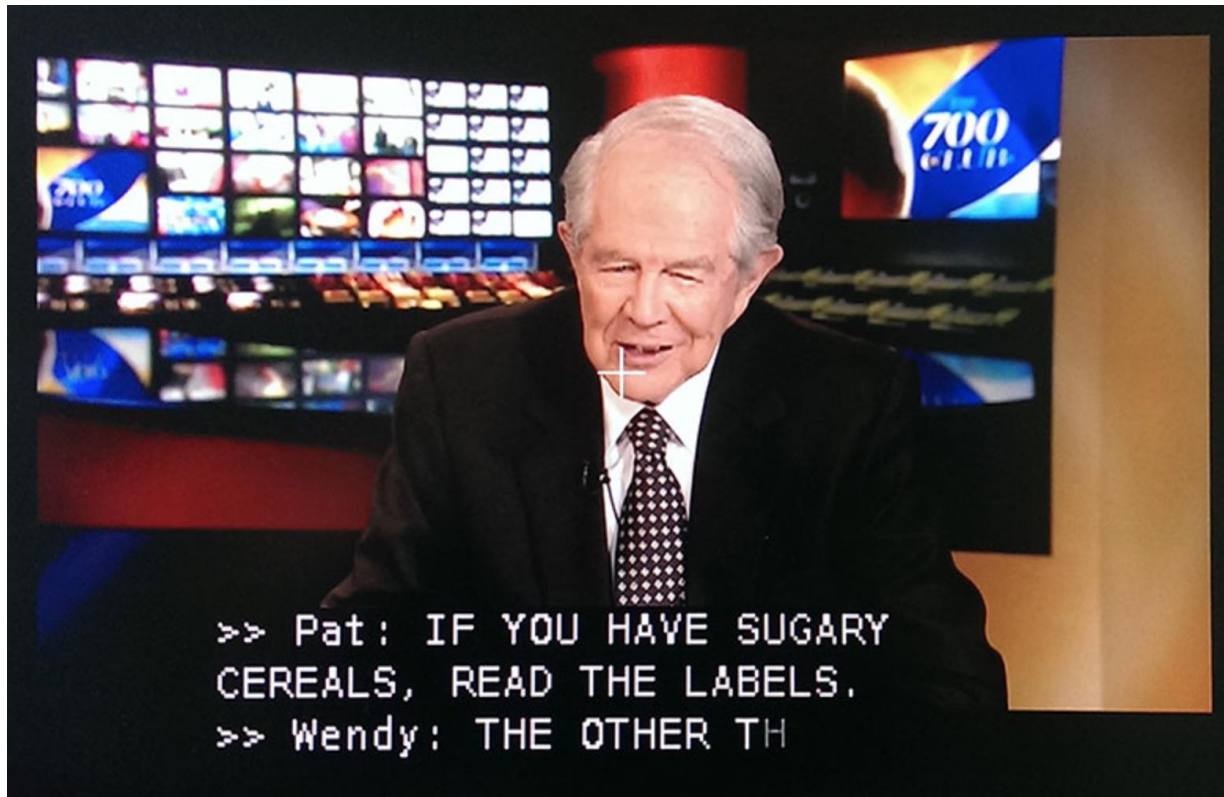
Curb Cut Phenomenon

- A design for people with impairments which benefits everyone
- Sidewalk curbs must be “cut” to allow wheelchair access
 - But benefits more than just wheelchair users
 - Easier to roll luggage, use strollers, carts, etc.
- Many technologies designed for those with disabilities have had this positive effect. More examples?



Curb Cut Example: Television Closed Captioning

- Text to describe to all significant audio content to hearing impaired viewers
 - What other uses does it provide?
In noisy environments (e.g., restaurants)



Curb Cut Example: Cassette Tape

- Cassette tapes were developed for a limited-market, and then *widely* adopted because of their portability
 - Developed as an alternative to reel-to-reel tape so visually impaired individuals could use books on tape more easily
 - Engineers didn't think average user would buy it because of inferior audio quality



Accessible User Interfaces

- Modern operating systems have a lot of built-in support
 - Set of end-user tools/capabilities
 - Set of toolkits to help create an application that is more universally accessible

Designed for you
Find products and tools based on your needs.
[Explore more tips and tools >](#)

**Vision**
Need a larger screen? A brighter screen? A narrator to read text? Find out about accessibility tools and features for people who are blind, color blind, or have low vision.
[Learn about vision-related tools >](#)

**Hearing**
For those who are hard of hearing, have hearing loss, or have deafness, our specialized features can provide solutions including closed captioning, mono sound, and live call transcription.
[Explore hearing-assistive tools >](#)

**Neurodiversity**
Innovative tools such as dictation and Windows Hello sign-in can make the digital world more accessible for those who with dyslexia, seizures, autism, or other cognitive differences.
[Discover tools for neurodiversity >](#)

**Learning**
Our applications for people living with learning disabilities can help increase focus, concentration, and understanding—and include tools to improve reading and writing skills.
[Uncover tools for learning >](#)

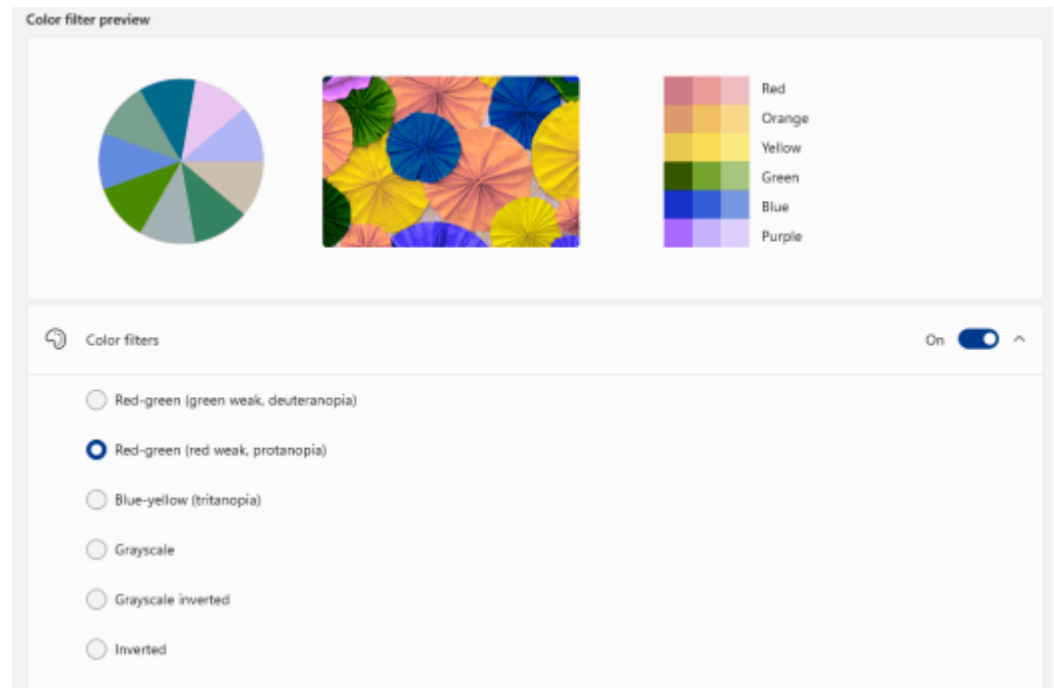
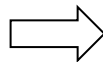
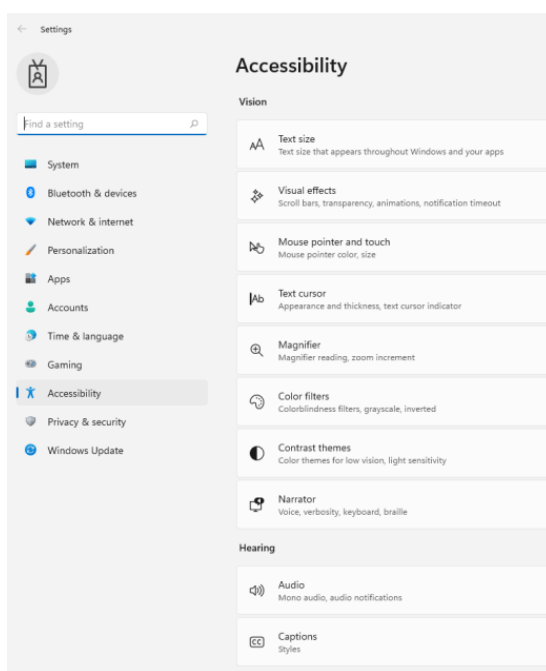
**Mobility**
Our suite of products helps people living with arthritis, quadriplegia, spinal cord injuries, and other mobility issues to navigate the digital world in new ways.
[Find mobility-assistive tools >](#)

**Mental health**
Learn more about assistive technologies for people living with issues such as bipolar disorder, anxiety, PTSD, depression, or ADHD. Our products can help reduce distraction and improve concentration and reading.
[Try mental health assistive tools >](#)

- <https://www.microsoft.com/en-us/accessibility>

Vision

- [Use color filters in Windows](#)
- [Change color contrast in Windows](#)
- [Make Windows easier to see](#)
- [Use Magnifier to make things on the screen easier to see](#)
- [Use color and contrast for accessibility in Microsoft 365](#)



Android accessibility overview

- [Use a screen reader with TalkBack](#)
- [Change your display](#)
- [Interaction controls](#)
- [Use a braille display](#)
- [Use Pixel's Magnifier](#)
- [Captions](#)
- [Audio](#)

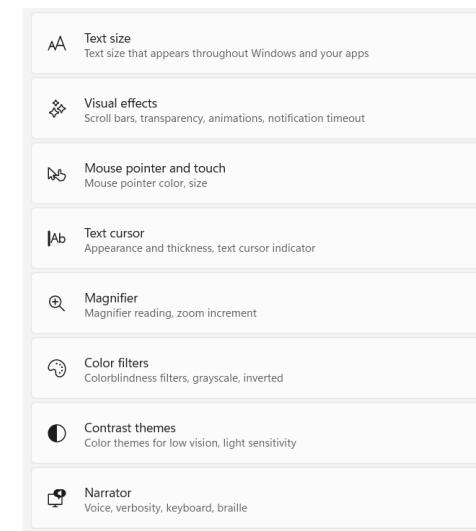
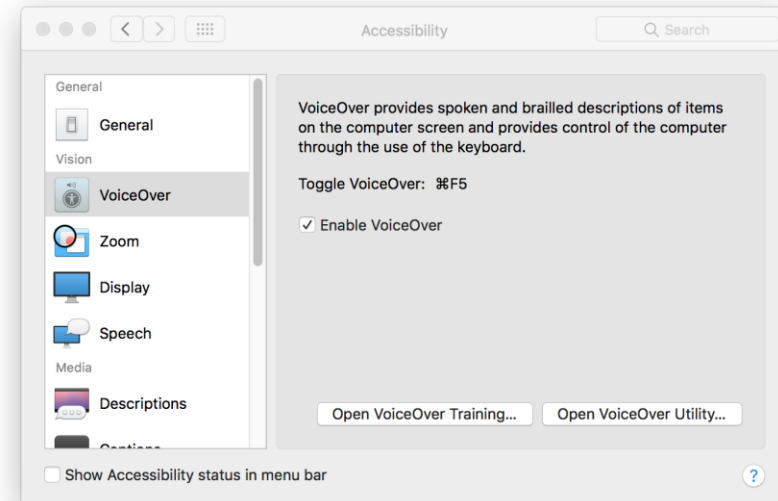
Use a screen reader with TalkBack



TalkBack

Operating System Support

- Typical support includes ability to:
 - Built-in screen readers
 - Ability to magnify portions of screen
 - Ability to increase contrast of screen
 - Ability to control interface using only keyboard
 - Ability to increase cursor size
- Fonts
 - People also adjust system-wide font size for minor visual impairments



Accessible User Interfaces

- Android and Windows provide hooks to integrate with accessibility functions
 - In Android: [Build more accessible apps](#)
- **Toolkits** provide features that allow you to provide additional information about your interface, individual components, and the functions they serve
 - This information allows accessibility software to expose your interface to users using different output modalities (e.g., screen readers)
 - Also allows accessibility software to control your software using alternative input methodologies (e.g., voice command)

Supporting Everyone: Some Examples

- Visual Impairment
- A visual impairment can limit one's ability to view text and graphical output
- With text output, a common solution is to use a screen reader
- What about with graphical output?

Visual Impairment

- There are a number of ways to provide alternative access to the information in graphical output including:
 - Providing **alternative text** for graphics so that a screen reader can be used (e.g., webpages)
 - Using **sound** to provide auditory cues regarding the graphical elements in the interface (e.g., mouse-over sounds for buttons)
 - Using **tactile output** to provide more information about graphics (Example: braille)

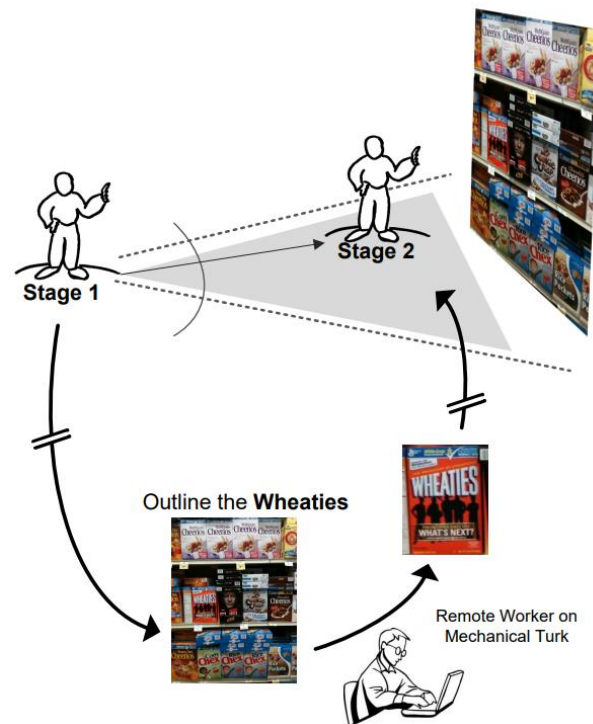


How Blind People Use Twitter & You Tube on the iPhone

- <http://youtu.be/c0nvdiRdehw>

VizWiz

- VizWiz enables blind people to take a picture and ask for assistance in finding a specific object.
- The request is first forwarded to remote workers who outline the object, enabling efficient and accurate automatic computer vision to guide users interactively from their existing cellphones



Curb Ramp Detection

- Remote crowd workers to find, categorize, and assess sidewalk accessibility problems using Google Street View (GSV) images
- Automatic Curb ramp detection using computer vision + crowdsourced verification



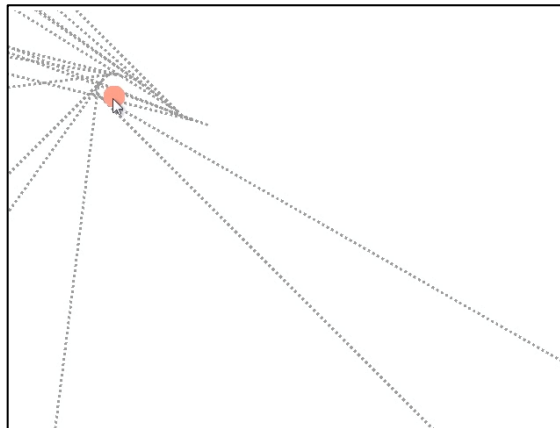
An Initial Study of Automatic Curb Ramp Detection with Crowdsourced Verification Using Google Street View Images

SUPPLE

- Interfaces should be personalized to better suit the contexts of individual users
- Automatically Generating Personalized User Interfaces
- Can generate user interfaces for people with motor and vision impairments
- Automatically generated, ability-based user interfaces significantly improve speed, accuracy and satisfaction of users with motor impairments compared to manufacturers' default interfaces
- Video link:
https://www.youtube.com/watch?v=B63whNtp4qc&feature=emb_title

Angle Mouse

- A pointing technique that improves the efficiency and ease of mouse pointing, especially for people with motor impairments
- Adjusts the mouse C-D gain based on the deviation of angles sampled during movement
- When angular deviation is low, the gain is kept high.
- When angular deviation is high, the gain is dropped
- Video link:
https://www.youtube.com/watch?v=O4ahGmHenps&feature=emb_title



Summary

It is important to design user interfaces to be not only usable, but accessible, such that the technology can reach more people, and have an impact across more situations.